Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Noise-Robust, Multi-Product Anomaly Detection Technology Embedded in an Industrial Automation Inspection Platform**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Optical Measuring Instrument	Industrial/ General Machinery	Industrial/General Machinery Software
Concept of Proposed Technology	In real manufacturing, vision-based quality inspection often faces two practical bottlenecks: (i) significant re-tuning cost whenever the product changes, and (ii) performance instability caused by real-world disturbances such as illumination variation, vibration, and environmental fluctuations. This project aims to deliver a general-purpose multi-product anomaly detection engine that can be embedded into an existing industrial automation and inspection software platform in a deployment-ready form. The proposed technology emphasizes: (i) multi-product generalization with minimal per-product rework, (ii) robust performance under illumination/vibration and other operational disturbances, (iii) uncertainty-/confidence-aware anomaly scoring and thresholding for practical operation, and (iv) modular APIs and operational logging/monitoring hooks for field deployment.			

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Rationale for Joint R&D Project	Korean manufacturing is facing rapidly increasing demand for automated inspection and quality assurance, driven by high-mix production and rising quality requirements. At the same time, workforce shortages and limited availability of highly skilled operators make conventional, labor-intensive inspection workflows and product-by-product customization costly and difficult to scale. In real production environments, inspection performance can also be destabilized by operational disturbances such as illumination changes, vibration, and other environmental fluctuations. As a result, general-purpose, out-of-distribution (OOD)-ready anomaly search that remains reliable under both product shifts and operational disturbances has become a critical enabler for practical, scalable industrial deployment. Our institute has strong expertise and state-of-the-art (SOTA) experience in OOD, anomaly detection, generalization, and robustness-oriented AI, which positions us to develop core source technology for multi-product anomaly search highest operational reliability. However, to translate this capability into tangible industrial value, success requires more than laboratory-level performance: it demands field validation under real operating constraints, system integration, and productization (packaging, deployment, and maintainability) that can only be achieved through close collaboration with an industrial partner. This joint R&D is therefore highly justified: it combines the institute's source-technology leadership with a Korean SME's commercialization and field deployment capabilities. The Korean SME will drive pilot operations, integration, and productization in real production settings, while the institute will deliver robustness- and generalization-centered methodology and rigorous quantitative verification. Together, the collaboration will enable a field-usable prototype and a clear commercialization roadmap within the three-year horizon. Ultimately, the resulting technology is expected to improve productivity, mitigate labor constraints, stabilize quality outcomes, and deliver a modular solution that can scale across industries—creating a broad positive impact on Korea's manufacturing competitiveness and digital transformation.
Key Words	1. Multi-Product Anomaly Detection 2. Noise Robustness 3. Uncertainty-aware scoring & thresholding 4. Deployment-ready API / modular integration 5. Industrial Validation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (8)
R&D Objectives	• **Develop a multi-product (general-purpose) anomaly search engine** ○ Enable deployment across diverse product families with minimal additional tuning, data rework, or re-training when products change. ○ Standardize the "minimum adaptation procedure" (configuration/calibration steps) to reduce on-site engineering overhead. ○ Reduce performance variability across product families by designing for generalization-first operation. • **Define real-world disturbance robustness and establish a standardized verification framework** ○ Systematically define manufacturing disturbances (e.g., illumination variation, vibration, environmental fluctuations) as stress-test scenarios. ○ Build a robust learning and evaluation framework covering data composition, validation scenarios, metrics, and reporting format. ○ Target not only peak performance but stable operational performance under changing conditions. • **Provide confidence/uncertainty-aware scoring and threshold operation for on-site use** ○ Deliver anomaly scoring with confidence/uncertainty signals to support practical decision-making on the shop floor. ○ Establish quantitative operating criteria for threshold management per product family. ○ Improve interpretability and operational control over the trade-off between false alarms and misses. • **Implement as platform-embeddable modules/APIs and complete pilot validation in real line conditions** ○ Package the core engine as deployment-ready APIs/modules compatible with an existing industrial automation/inspection platform. ○ Integrate operational capabilities such as logging, monitoring, and reporting. ○ Conduct pilot validation under real operating constraints to verify both performance and operability. • **Deliver a commercialization-ready prototype and technology transfer package** ○ Provide a field-usable prototype along with installation/operation guides, evaluation protocols, integration documents, and maintenance/update guidelines. ○ Enable sustained use and future expansion by the partner through training and technology transfer materials. ○ Ensure the final outputs are suitable for productization rather than one-off demonstrations.			
R&D Target Specifications	• **Multi-product validation** ○ Validate performance on ≥3 distinct product groups that differ in appearance/process/operating conditions. ○ Quantify product-to-product performance variance using a unified evaluation protocol to ensure comparability and reproducibility. ○ Define and track the "adaptation cost vs. performance" relationship (e.g., how much extra work is needed when switching products).			

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	• **Robustness under operational disturbances**
	  ○ Establish stress-test conditions reflecting illumination variation (brightness, color temperature, shadowing, reflections) and vibration/environmental fluctuations (micro-shake, input perturbations, sensor/fixture shifts).
	  ○ Constrain performance degradation under stress tests using jointly agreed criteria such as:
	  ▪ performance retention vs. baseline
	  ▪ stability/variance across repeated runs
	  ▪ sensitivity to drift over time
	  ○ Ensure stress tests are reproducible and practical for field execution.
	• **False-alarm control**
	  ○ Define and achieve a target False Alarm Rate suitable for line operation.
	  ○ Manage not only the average rate but also variability across time and operating conditions to reduce operational disruption.
	  ○ Provide case-level reporting (representative examples, confidence indicators, triggering conditions) to support fast root-cause analysis**.**
	• **Deployment requirements**
	  ○ Package the solution to meet platform constraints (cycle time, compute resources, network/storage, operational policies) for real-time or near-real-time use.
	  ○ Provide versioning and compatibility principles to support safe updates and rollbacks.
	  ○ Include deployment documentation to minimize integration risk and operational burden.
	• **Operability**
	  ○ Ensure consistent linkage of anomaly scores, thresholds, and operating conditions to logging/monitoring/reporting pipelines.
	  ○ Provide a reproducible evaluation protocol detailing data splits, scenarios, and metric computation.
	  ○ Offer a periodic health-check procedure (sampling plan, reporting cadence, alert criteria) to detect performance drift during operation.

R&D Plan	**R&D Plan *(Total 3 years: 0.5 year + 2.5 years)***
	
	Phase 1 (Pre-planning, 6 months): Finalize requirements, data, and evaluation criteria
	
	• **Refine requirements and success criteria**
	  ○ Confirm multi-product scope (product group definition, switching frequency, operating scenarios).
	  ○ Formalize operational constraints (cycle time, throughput, allowable false alarms, operating workflow).
	  ○ Agree on performance targets including both model metrics and operational metrics.
	• **Define data and operational conditions; establish governance**
	  ○ Agree on data scope, labeling policy, and representativeness requirements.
	  ○ Establish dataset governance: versioning, metadata, quality checks, and access/security policies**.**
	  ○ Enumerate likely sources of variation (environment/equipment/operator factors) and incorporate them into evaluation.
	• **Build disturbance reproduction and stress-test protocols**
	  ○ Standardize disturbance scenarios (illumination variation, vibration, environmental fluctuation).
	  ○ Define test severity bands, repetitions, and reporting templates.
	  ○ Agree on baseline vs. stressed performance criteria (retention and stability metrics).
	• **Align platform integration approach**
	  ○ Finalize API specifications, I/O formats, exception handling, and monitoring hooks.
	  ○ Align UI/UX expectations for operators (alerts, reports, case review).
	  ○ Confirm deployment environment assumptions (edge/on-prem), update/rollback policy, and operational permissions.

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Phase 2 (Main research, 2.5 years): Development, integration, and field validation — Year-by-Year Plan

Year 1 (Months 7–18): Core Engine Development + Baseline Integration

- **Generalization & Robustness Core Development**
  - Develop the first production-oriented baseline of the multi-product anomaly search engine.
  - Establish an initial robustness training/evaluation pipeline aligned with Phase 1 stress-test protocols.
  - Identify dominant failure modes across product groups and disturbance scenarios (e.g., false alarms driven by illumination/vibration changes).
  - Implement first-round improvements (including statistical components where appropriate) and validate via repeatable internal benchmarks.
- **Confidence/Uncertainty-Aware Scoring**
  - Introduce confidence/uncertainty-aware scoring outputs and define how they map to operational decisions.
  - Create an initial thresholding strategy and reporting format for on-site review (e.g., score distributions, representative cases, confidence cues).
- **Integration Prototype**
  - Implement a platform-embeddable API/module (v1) with stable I/O, exception handling, and logging hooks.
  - Deliver an early integration demo in a controlled environment and confirm runtime feasibility under the target operational constraints.

Year 1 Deliverables

- Engine baseline (v1) + standardized evaluation report
- Stress-test results (baseline vs. disturbed) + failure-mode analysis
- API/module (v1) + integration guide (draft)
- Confidence/uncertainty scoring (v1) + thresholding/reporting draft

Year 2 (Months 19–30): Robustness Hardening + Multi-Product Expansion + Pilot-Ready Release

- **Robustness Hardening & Stress-Test Closure**
  - Systematically harden robustness by iterating: weak-condition discovery → mitigation → re-validation.
  - Expand stress-test coverage to include “combined” disturbance conditions and operational drift-like variations.
  - Define a robustness acceptance gate (pass/fail criteria) agreed in Phase 1 and apply it as a release requirement.
- **Multi-Product Scaling & Adaptation Cost Reduction**
  - Expand validation coverage across ≥3 product groups, prioritizing diversity of operating conditions.
  - Reduce per-product adaptation effort by formalizing a minimum adaptation kit (configuration, calibration checks, data sampling recipe).
  - Quantify and report “adaptation cost vs. performance” to demonstrate scalability.
- **Operational Scoring & Thresholding**
  - Refine confidence/uncertainty-aware scoring to improve stability and interpretability.
  - Deliver a threshold governance workflow**:** how thresholds are set, reviewed, updated, and audited per product group.
  - Add case-based analytics to support operator decisions and maintenance cycles.

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- **Platform Module/API + Operability Features**
  - Upgrade to **API/module (v2)** focusing on performance stability, observability, and operational safety:
    - structured logging, health checks, monitoring metrics, and alert triggers
  - Provide deployment artifacts (container/package) and expand documentation for install/config/operate/update/rollback.

Year 2 Deliverables

- Engine (v2) meeting robustness acceptance gate (pilot-ready)
- Multi-product validation report (≥3 product groups) + adaptation cost analysis
- Scoring/threshold governance specification + operator-facing reporting format
- API/module (v2) + deployment package + operability (monitoring/logging) integration

Year 3 (Months 31–36, first half-year): Field Pilot + Reliability Evaluation + Technology Transfer & Commercialization Package

- **Pilot Deployment & On-site Validation**
  - Execute pilots across multiple product groups under real operating constraints.
  - Validate both performance **and** operability (stability, failure recovery, monitoring, logging completeness).
  - Establish an on-site feedback loop and triage process for issues.
- **Reliability Evaluation & Continuous Improvement**
  - Perform reliability evaluation focusing on:
    - robustness under recurring disturbances
    - drift sensitivity and stability over time
    - false alarm variability and miss patterns
  - Apply targeted fixes and conduct re-validation under the same standardized protocol.
- **Technology Transfer Package & Commercialization Roadmap**
  - Deliver a complete transfer package including:
    - core module/API, interface specs, deployment artifacts
    - reproducible evaluation protocol and reporting templates
    - operation manual(install/operate/monitor/update/rollback), troubleshooting playbook
    - training materials and handover workshops
  - Provide a commercialization roadmap:
    - target use-cases, deployment prerequisites, scaling plan across additional product lines
    - maintenance/update plan and future improvement directions

Year 3 Deliverables

- Pilot validation report + reliability evaluation report (field-ready evidence)
- Final release candidate (v3) + stable integration package
- Full technology transfer kit + training completion record
- Commercialization roadmap + scaling plan

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Develop source technology for multi-product anomaly search and robustness-centered learning/evaluation methodology • Define standardized stress-test protocols, quantitative metrics, and verification reporting • Design uncertainty-/confidence-aware anomaly scoring and thresholding for stable operation • Provide reference implementation, API specifications, and joint debugging support for platform integration • Deliver documentation and training for technology transfer, including reproducible evaluation and deployment guidance
Expected R&R of Korean SME Partner	• Provide and operate an existing industrial automation and inspection software platform with field deployment experience • Provide access to multi-product production environments and support continuous data collection and pilot operations • Support controlled reproduction/provision of disturbance conditions such as illumination variation and vibration for verification • Execute productization activities: integration engineering, UI/UX for operators, logging/monitoring, maintenance and release management • Provide a commercialization plan including customer validation and scaling roadmap after successful pilots

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• **A reusable, general-purpose anomaly search engine and operational protocol for high-mix production**
	  ○ Deliver a multi-product anomaly search engine designed for reuse across changing product families.
	  ○ Provide standardized operating procedures, including metrics, threshold operation, reporting templates, and periodic health checks.
	  ○ Enable scalable inspection workflows that reduce reliance on one-off, product-specific customization.
	• **A robustness-oriented inspection module that maintains stable performance under real-world disturbances**
	  ○ Establish standardized stress tests reflecting illumination variation, vibration, and environmental fluctuations.
	  ○ Deliver a module that prioritizes operational stability, reliability, and reproducibility over fragile one-time demo performance.
	  ○ Improve resilience against unavoidable day-to-day variations in production environments.
	• **A field-deployable prototype embedded into the target platform via APIs/modules with logging/monitoring integration**
	  ○ Provide platform-embeddable APIs/modules and connect them to operational logging/monitoring/reporting.
	  ○ Offer case-level reporting (representative examples, confidence indicators, triggering conditions) to support fast debugging and continuous improvement.
	  ○ Deliver an implementation that can be piloted under real production constraints.
	• **Lower re-tuning cost during product switching and improved operational efficiency through reduced false alarms/misses**
	  ○ Reduce the engineering burden associated with product changeovers by minimizing per-product tuning and re-validation.
	  ○ Decrease operational disruptions caused by false alarms and improve quality stability by limiting misses.
	  ○ Contribute to higher throughput, better line uptime, and more predictable quality outcomes—particularly valuable under persistent labor shortages.
	• **Technology-transfer-centered joint outcomes, enabling Korean-side ownership and sustained internalization**
	  ○ Beyond joint research, the project explicitly targets technology transfer so that the Korean side can internalize and expand the results.
	  ○ Deliverables will include transferable assets such as core modules, interface specifications, reproducible evaluation protocols, deployment/operation documentation, and training materials.
	  ○ The Canadian university will support the Korean partner/institution in taking ownership of the developed IP/know-how necessary for practical deployment and productization, with the detailed scope and terms to be finalized through consortium agreements and contracting.
	• **A positive spillover effect: advancing inspection capability across Korean manufacturing sectors**
	  ○ The general-purpose engine and standardized verification framework are designed to be extendable to other inspection use cases.
	  ○ Promote best practices for standardized evaluation, reliability-focused deployment, and data-driven continuous improvement.
	  ○ Strengthen the domestic ecosystem by elevating the baseline capability of industrial inspection technology.
	• **Global market-oriented joint R&D outcomes and strengthened commercialization potential**
	  ○ High-mix, disturbance-robust inspection technology is broadly demanded in global manufacturing markets.
	  ○ The project establishes a foundation for scalable, modular commercialization pathways, including broader deployments, new customer validation, and future product line expansion.

	○ Improve Korea's manufacturing competitiveness by enabling faster adoption of reliable inspection automation.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization